Data Collection Methods

Carlos Manuel Herrera Castillo



University of Zurich[UZH]

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From research question to data

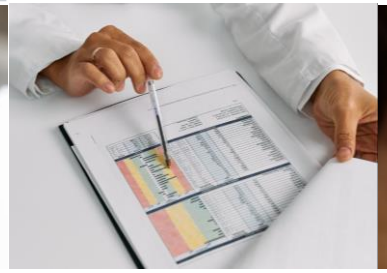
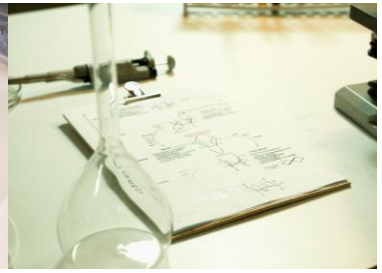


“Does physical activity improve sleep quality among university students?”

- Physical activity measurement
- Sleep measurement
- Characteristics of participants
- Perhaps repeated measurements over time

Research question → What do we need to observe? → How will we collect it?

Common data collection methods

Devices / sensors	Laboratory / clinical measurements	Existing records	Surveys / questionnaires	Observations	Existing secondary data
• Movement • Heart rate • Sleep • Glucose • Environmental exposure	• Biomarkers • Physiological parameters • Diagnostic measures	• Clinical history • Diagnoses • Treatments • Healthcare utilization	• Behaviours • Symptoms • Attitudes • Perceptions • Quality of life	• Behaviours • Interactions • Clinical practice • Environments	• Trends • Meta-analysis
					

The same phenomenon can be measured differently



Take **physical activity**:
Questionnaire → self-reported activity
Accelerometer → objectively recorded movement
Interview → experience and perception of physical activity
Observation → observed behaviour

Are these measuring exactly the same thing?
No.

They provide **different operationalizations of the underlying concept**.

The method influences the data we obtain.

Match the method to the question

How do patients experience X?	Survey
Does treatment X change biomarker Y?	Interview
What happens to patients after discharge?	Observation
How common is symptom X?	Laboratory measurement
Does intervention X change physical activity?	Device/sensor
How do clinicians behave in situation X?	Clinical records / longitudinal follow-up



There is no universally “best” data collection method.

The appropriate method depends on:

- The question
- The study design
- The research paradigm
- Feasibility
- Ethics
- Resources

Research paradigms



Quantitative

How much? How many? Is X associated with Y? Does X cause Y?

Common methods:
Surveys
Measurements
Experiments
Sensors
Laboratory tests



Qualitative

How? Why? How is something experienced or understood?

Common methods:
Interviews
Focus groups
Observation
Textual/document analysis

Paradigm → question → type of evidence → appropriate methods

Context and feasability

“What could make your ideal data collection method impossible?”



Time
Can we follow participants for 12 months?

Logistics
Can samples be collected and stored correctly?

Cost
Can we provide every participant with an expensive device?

Data quality
Will participants answer accurately?

Access
Can we recruit the required population?

Participant burden
Will participants complete a 60-minute questionnaire every week?

Ethics
Can we collect this information?

Publication exercise

METHODS

The Cleveland Veterans' Affairs Medical Center is a 210-bed acute-care facility. At the time of the study, all hospitalized patients received alcohol-based hand sanitizer supplied with other toiletry items, but no education on patient hand hygiene. Supplemental Table 1 provides detailed information on the study facility.

The Cleveland Veterans' Affairs Medical Center's institutional review board approved the study protocol. From October through March 2014, we conducted a non-blinded parallel randomized trial of a patient hand-hygiene intervention versus standard care for a convenience sample of patients newly admitted to 4 medical-surgical wards with anticipated lengths of stay of at least 2 days. Supplemental Table 2 provides details on study procedures. Patients were excluded if they had dementia, difficulty performing hand hygiene, or methicillin-resistant *Staphylococcus aureus* (MRSA) colonization or *Clostridium difficile* infection. Verbal consent was obtained. Patients were randomized based on a coin toss. For patients in the intervention group, education was based on a "Four Moments for Patient Hand Hygiene" model.[7] In addition to a "Four Moments for Patient Hand Hygiene" poster (Supplemental Figure 1), intervention patients received illustrations of the effectiveness of alcohol-based hand sanitizer (Supplemental Figure 2).[8] A bottle of alcohol-based hand sanitizer was placed at the bedside, and intermittent soap-and-water hand washing was encouraged. Patients received re-education during daily follow-up visits. During each visit, research personnel directly facilitated the use of alcohol-based hand sanitizer after the collection of hand cultures. Patients randomized to the control group had their hands cultured, but they did not receive education or hand sanitizer.

Premoistened BD BBL CultureSwabs (Becton Dickinson, Cockeysville, MD) were used to culture patients' hands upon admission and on hospital days 2, 4, and 5 to detect important healthcare-associated pathogens that are susceptible to alcohol-based hand sanitizer: MRSA, vancomycin-resistant Enterococci (VRE), and fluoroquinolone-resistant gram-negative bacilli. The entire surface area of both hands was sampled. To test whether hand hygiene might impact environmental contamination, 5-cm × 10-cm areas of the bed rail and bedside table and the entire surface area of the call button were sampled using swabs at the same time points. The swabs were cultured for MRSA and VRE as previously described.[9] For fluoroquinolone-resistant gram-negative bacilli, the swabs were plated on MacConkey agar (Becton Dickinson, Cockeysville, MD) containing 10 μg/mL ciprofloxacin. The microbiologist processing the cultures was blinded to the study group.

Information on medical conditions, medications, devices, mobility, and long-term care facility residence was obtained through chart review; 3 months after discharge, medical records were reviewed to assess for new colonization or infection with the pathogens. The primary outcome we investigated was new acquisition of hand contamination with pathogens. Based on preliminary data showing that ~35% of patients acquired hand contamination, a power calculation indicated that 45 patients per group would provide 80% power to detect a reduction from 35% to 10% acquisition of hand contamination. Bivariate analyses were conducted to compare characteristics of groups. Fisher's exact test was used for categorical data and the Student paired *t* test was used for normally distributed data. Data were analyzed using SPSS statistical software version 10.0 (SPSS, Chicago, IL).

Sunkesula, et al., 2017

Please identify:

1. What was the research question?
If it isn't explicitly stated, infer it.
2. What data were collected?
3. How were the data collected?
4. Why do you think this method was chosen?
5. What are the main strengths?
6. What are the main challenges or sources of bias?

What should good data collection reporting tell us?



Who?
Who provided the data?

What?
What was measured?

How?
Which instrument/procedure was used?

When?
At what time/frequency?

Where?
In what setting?

Under what conditions?

How was data quality controlled?

Take home messages

A good research question determines what evidence we need.
The study design determines how we can obtain that evidence.
The data collection method must therefore be appropriate to:
Question + Design + Paradigm + Context
And one final sentence:
Good data collection is not simply collecting more data; it is collecting the right data, in the right way, to answer the question.

